

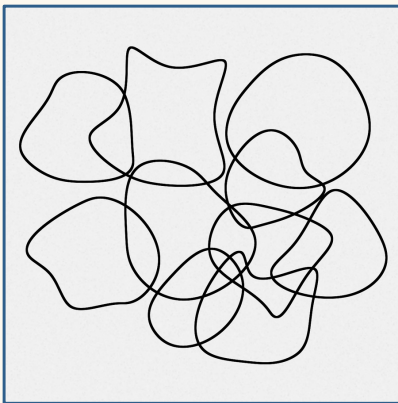
PHASE A • PROCEDURAL GENERATION

1 • creation.py

```
# creation.py (deterministic)
def generate(seed):
    rng = default_rng(seed)
    n = rng.integers(10, 16) # GT
    loops = [closed_loop(rng)
              for _ in range(n)]
    return loops, n
```



2 • scaffold



PHASE B • PHOTOREALISTIC RE-RENDERING

3 • GPT-image-2 prompt

Re-render as a photo-realistic top-down photo of ropes tangled on a weathered surface... Each loop follows the EXACT same path, with proper over or under crossings... Preserve every loop; add or remove none.



4 • photo



How many distinct closed loops are tangled together?

GROUND TRUTH

$n = 10$

GEOMETRY PRESERVED

TANGLED LOOP COUNTING

RECOVERED FROM PHOTO

$= 10$